5 Original benchmark reassessed

Table 1 uses the four evaluation sentences, with 16 cases per noise and method. The SNRs are unchanged; all STOI values use the common retrospective rescore. Both enhanced methods raise average SNR over the input, but the incremental gain from updating is small.

Table 1 Original evaluation means with standard STOI

Noise	Method	SNR dB	STOI
Car	Input	7.103	0.7968
Car	Fixed	10.814	0.7966
Car	Original VAD	10.863	0.7894
Street	Input	7.207	0.8236
Street	Fixed	9.656	0.8034
Street	Original VAD	9.800	0.8023

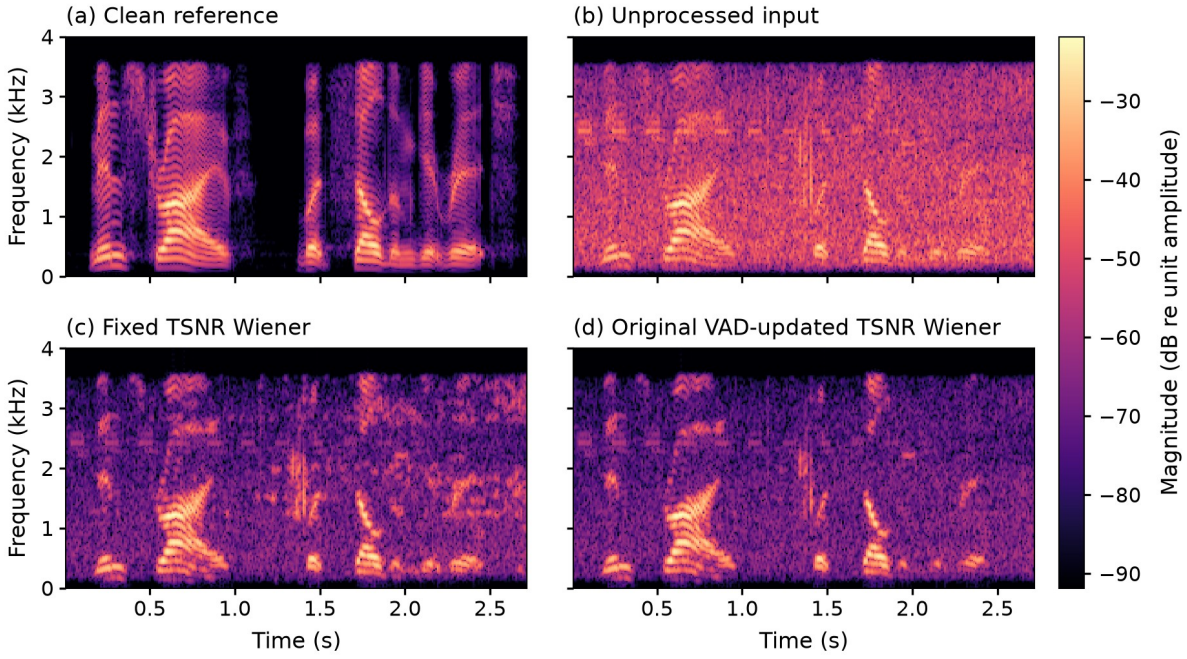


Figure 1 sp06 in street noise at nominal 5 dB. The clean, input, fixed and original VAD panels share the analysis settings and 70 dB colour range. These spectra are unchanged by the metric rescore.

For this example, original VAD raises SNR from 9.793 to 9.897 dB relative to fixed estimation, but STOI falls from 0.7539 to 0.7424. The darker background accompanies lower waveform error without higher predicted intelligibility. In the corresponding trace, updates overlap 42 of 166 clean speech-energy proxy frames; this is a diagnostic clue, not proof of the cause of every score change.

Across the evaluation set, original VAD changes mean SNR by +0.049 dB in car and +0.144 dB in street noise. Mean STOI changes by -0.00717 and -0.00103. It improves SNR in 18/32 cases, STOI in 8/32, and both in 5/32. The small mean SNR advantage therefore cannot establish a generally better enhancer.

7 Condition effects and sentence sensitivity

Table 4 separates selected-minus-fixed effects across the eight noise/SNR groups, each containing the same six sentences. A positive group average need not imply consistent improvement for individual sentences.

Table 4 Selected VAD minus fixed estimation

Noise	Nominal SNR dB	SNR change dB	STOI change	Both improve
Car	0	+0.0675	-0.00122	1/6
Car	5	+0.0181	-0.00015	3/6
Car	10	+0.0165	+0.00029	2/6
Car	15	+0.0358	+0.00064	4/6
Street	0	-0.0849	-0.00003	1/6
Street	5	+0.3316	-0.00596	0/6
Street	10	+0.1022	+0.00207	3/6
Street	15	+0.0219	-0.00079	0/6

The largest mean SNR gain, street at 5 dB, coincides with a STOI loss and no sentence improving both scores. Street at 10 dB improves both averages but only three of six sentences improve both. Different source excerpts can occur at different nominal SNRs, so these rows do not isolate the effect of SNR alone.

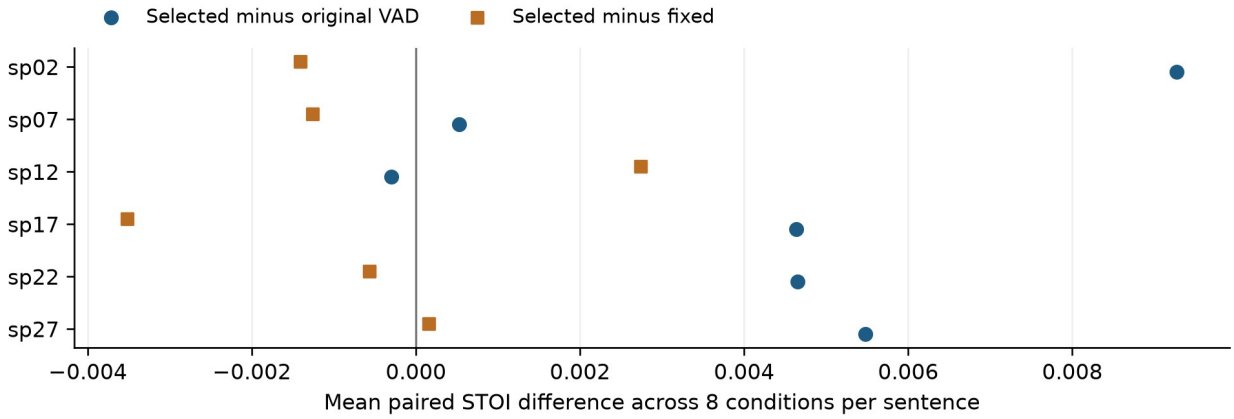


Figure 2 Every sentence-level mean paired STOI effect. Each point averages eight conditions. The six sentences, rather than the 48 repeated conditions, are the units of the sensitivity check.

Five of six sentence means favour selected over original VAD. Removing any one sentence leaves a positive mean STOI effect of +0.00300 to +0.00491. Only two sentence means favour selected over fixed; all leave-one-out averages remain negative (-0.00132 to -0.00007). The direction is not driven by one sentence, although these checks do not quantify population uncertainty.

The predetermined sp02/street/5 dB demonstration remains a counterexample: selected VAD reduces proxy overlap from 25/193 to 10/193 frames versus original VAD, yet STOI falls from 0.7278 to 0.7165. Lower proxy overlap alone therefore does not guarantee a better intelligibility score.